Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The Flood Prediction System uses Machine Learning and historical weather data to predict flood risk. It processes user inputs using a trained XGBoost model and displays the prediction through a Flask web application. The system requirements are categorized into Functional Requirements (FR) and Non-Functional Requirements (NFR).

- **Functional Requirements (FR):** Define the core functionalities of the Flood Prediction System, including data input, validation, prediction generation, result display, and record management. These describe what the system should do.
- **Non-Functional Requirements (NFR):** Define the quality attributes of the Flood Prediction System, such as performance, accuracy, reliability, usability, scalability, and maintainability. These describe how the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Authorized users can securely access the flood prediction system.	High
2	Authorization levels	Disaster management authorities and analysts can access prediction features	High
3	External interfaces	Connects with weather datasets and supports future weather API integration.	Medium

4	Transactions processing	Accepts weather parameters and generates flood prediction results.	High
5	Reporting	Displays flood prediction results and model accuracy reports.	Medium
6	Business rules, etc.	Predicts flood risk using Decision Tree, Random Forest, KNN, and XGBoost models.	High
7	Compliance to laws or regulations	Handles weather data according to applicable data privacy guidelines.	Medium
8	*Other*	Supports deployment on IBM Cloud for remote access and scalability.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	Generate flood prediction quickly.	Prediction within 2 seconds
2	Scalability	Support multiple users and datasets.	Supports cloud deployment
3	Security & Data Privacy	Protect user input and prediction data.	Secure access and data protection
4	Reliability & Availability	Ensure continuous operation with accurate predictions.	99% system availability

5	Usability & Accessibility	Simple and easy-to-use web interface.	User-friendly interface
6	Other	Easy maintenance and future model updates	Modular and maintainable system